MECHENG 730/736
QUESTION/ANSWER BOOKLET

THE UNIVERSITY OF AUCKLAND

SEMESTER TWO, 2025
Campus: City

MECHANICAL ENGINEERING

Biomechatronics Systems

(Time Allowed: TWO Hours)

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

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INSTRUCTIONS FOR THIS EXAM ARE PROVIDED ON THE NEXT PAGE.

NOTE

1. This exam is worth 50% of your final grade.
2. It will be marked out of 100 marks.
3. There are 3 parts. Part 1 and 3 are worth 30 marks each and part 2 is worth 40 marks.
4. You must answer ALL questions.
5. Answer each question in the space provided below the question.
6. If you require additional answer pages, overflow pages are provided at the end of this booklet. Within the designated space for the question you answer, you must provide a clear indication that your answer is continued on an overflow page e.g., “Answer is continued on overflow page X”.
7. Write neatly and legibly. Use a dark ink that does not show through the paper.
8. Cross out notes and other work you do not wish to have assessed.

DO NOT REMOVE ANY PAGES FROM THE QUESTION / ANSWER BOOKLET.

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PART 1: EEG and sleep (30 marks)

Each of the following multiple-choice questions (questions 1 to 20) is worth 1 mark. For each question circle the one best answer.

1. Which of the following sleep features is most characteristic of narcolepsy?

A. Sleep-onset REM periods (SOREMPs)

B. Prolonged latency to NREM stage 2 sleep

C. Complete absence of REM sleep

D. Excessive SWS rebound on recovery nights

2. Which one of the following statements best represents the current scientific thinking on the relationship between sleep and learning?

A. Sleep plays little role in learning or memory consolidation.

B. Only REM sleep, not NREM sleep, has been linked to learning.

C. Both REM and NREM sleep play an important role in learning and memory consolidation.

D. The association between sleep and learning is universally accepted scientific consensus.

3. Which of the following statements best describes the prevalence of obstructive sleep apnea (OSA)?

A. OSA is equally prevalent in men and women across all ages.

B. OSA prevalence is higher in women than in men, particularly at younger ages.

C. OSA prevalence increases with age and is about twice as common in men as in women, though the sex difference diminishes in older adults.

D. OSA prevalence is unrelated to obesity or body mass index.

4. Which of the following questionnaires is specifically designed for preoperative screening of OSA risk?

A. Berlin Questionnaire

B. STOP-Bang questionnaire

C. Epworth Sleepiness Scale

D. Pittsburgh Sleep Quality Index

5. What is the primary cause of excessive sleepiness in individuals with OSA?

A. Increased sympathetic activity

B. Sleep fragmentation due to repeated arousals

C. Chronic hypercapnia

D. Elevated catecholamine levels

6. Which of the following is an example of an unusual behavior that can occur during sleep?

A. Sleepwalking and teeth grinding

B. Increased heart rate and blood pressure

C. Enhanced perceptual awareness of the environment

D. Permanent muscle paralysis

7. Which of the following statements best reflects the relationship between circadian rhythms and REM sleep?

A. REM sleep is independent of circadian processes.

B. REM sleep episodes are distributed toward the last third of the night, linked to circadian oscillations.

C. REM sleep occurs only in the first third of the night when circadian temperature is lowest.

D. REM sleep is equally distributed throughout all sleep cycles, regardless of circadian influence.

8. What does the apnea–hypopnea index (AHI) measure?

A. The number of apneas and hypopneas per night of sleep

B. The number of apneas and hypopneas per hour of sleep or recording

C. The number of oxygen desaturation episodes per night

D. The total minutes spent in apnea during sleep

9. Which of the following electroencephalogram (EEG) features is characteristic of NREM sleep?

A. Desynchronized low-voltage fast activity

B. Synchronous patterns, including K-complexes, and high-voltage slow waves

C. Absence of any electrical activity in the brain

D. Continuous REM bursts

10. Which EEG change is most often taken as the hallmark of sleep onset?

A. Transition from alpha rhythm to a low-voltage, mixed-frequency pattern (stage 1 sleep)

B. Continuous rhythmic alpha (8–13 cps) activity in the occipital region

C. Sudden onset of high-voltage slow waves typical of stage 4 sleep

D. Appearance of REM bursts and PGO waves

11. Which of the following is commonly associated with sleep fragmentation and frequent arousals?

A. Sleep disorders such as sleep apnea, or upper airway resistance syndrome

B. Jet lag and circadian rhythm shifts

C. Exclusive suppression of REM sleep in healthy young adults

D. Increased melatonin secretion at sleep onset

12. Successful treatment of sleep apnea syndromes (e.g., with CPAP) often results in which of the following?

A. Permanent suppression of REM sleep

B. Immediate disappearance of NREM stages 3 and 4

C. Large rebounds of SWS or REM sleep

D. Increased frequency of SOREMPs

13. Which of the following is a notable feature of slow wave sleep (SWS) in young children?

A. It is qualitatively similar to that of older adults and easily disrupted

B. It is diminished by age 10, with no measurable difference from adults

C. It is intense, sometimes preventing arousal even with very loud stimuli

D. It typically accounts for less than 5% of their total sleep time

14. According to the two-process model of sleep, which factor primarily determines the distribution of slow-wave sleep at the beginning of the night?

A. Circadian rhythms synchronized to body temperature oscillations

B. Volitional control of sleep timing

C. Homeostatic sleep pressure related to the length of prior wakefulness

D. Genetic determinants of sleep need

15. Which of the following best distinguishes the two primary states of sleep?

A. They are defined entirely by behavioural observations.

B. They are based exclusively on dream activity.

C. They occur only in humans and primates.

D. They are rapid eye movement (REM) and non–rapid eye movement (NREM) states, distinct from each other and from wakefulness.

16. Which of the following best describes sleep in newborn infants?

A. Sleep is entered through NREM stage 1, with a cycle period of about 90 minutes

B. Sleep is often entered through REM sleep, with a cycle period of about 50–60 minutes

C. Sleep is consolidated into a nocturnal pattern immediately after birth

D. Fully developed EEG patterns of NREM stages are present from birth

17. In a healthy young adult, which sleep stage typically occupies the largest proportion of total sleep time across the night?

A. Stage 1 sleep

B. Stage 2 sleep

C. Slow wave sleep

D. REM sleep

18. Which surgical approach has shown the largest average reduction in apnea–hypopnea index (AHI) across multiple studies of OSA patients?

A. Uvulopalatopharyngoplasty

B. Tongue base reduction

C. Maxillomandibular advancement

D. Tracheotomy

19. Which of the following is considered one of the most notable findings about sleep in elderly people?

A. A uniform pattern of reduced REM sleep across individuals

B. A profound increase in interindividual variability of sleep patterns

C. A universal decline in stage 2 sleep percentage

D. A complete absence of arousals during sleep

20. How do the four stages of NREM sleep relate to arousal thresholds?

A. Arousal thresholds are highest in stage 1 and lowest in stage 4.

B. Arousal thresholds are lowest in stage 1 and highest in stage 4.

C. Arousal thresholds remain constant throughout NREM.

D. Arousal thresholds fluctuate randomly across stages 1–4.

21. This question is about the following Matlab script:

```
 1  N_OBSV_PER_SAMPLE = 20;
 2  [x1,y1] = fnGetSample(N_OBSV_PER_SAMPLE, [-1 0], [[1 0];[0 1]]);
 3  [x2,y2] = fnGetSample(N_OBSV_PER_SAMPLE, [-4 -4], [[1 0];[0 1]]);
 4
 5  % Plot all observations in the 2D feature space
 6  figure; hold on
 7  plot(x1,y1, 'or')
 8  plot(mean(x1),mean(y1), 'ok', 'MarkerFaceColor', 'black', 'MarkerSize', 4)
 9  plot(x2,y2, 'ob')
10  plot(mean(x2),mean(y2), 'ok', 'MarkerFaceColor', 'black', 'MarkerSize', 4)
11  axis square
12  axis([-10 10 -10 10])
13  set(gca, 'xtick', -10:5:10, 'ytick', -10:5:10)
14  grid on
15  xlabel('Feature 1 (a.u.)')
16  ylabel('Feature 2 (a.u.)')
17
18  % Train classifier using observations generated above
19  training = [[x1;x2] [y1;y2]];
20  group = [zeros(N_OBSV_PER_SAMPLE,1); ones(N_OBSV_PER_SAMPLE,1)];
21  obsv_test = [1 1; 0 -8; 0 0; -6 0; -5 -5];
22  CLASSIF_TYPE = 'diaglinear';
23  [CLASS,~,~,~,COEF] = classify(obsv_test,training,group,CLASSIF_TYPE);
24  disp(CLASS)
25
26  % Get model coefficients and plot decision boundary
27  K = COEF(1,2).const;
28  L = COEF(1,2).linear;
29  f2 = @(x) (-K - L(1)*x) / L(2);
30  fplot(f2, [xlim], 'LineStyle', '-', 'Color', 'black');
31
32  function [x,y] = fnGetSample(n, mu, S)
33  % This function returns observations drawn from a
34  % 2-dimensional normal distribution. The number of
35  % observations is n, the normal's position is given by mu,
36  % and the covariance matrix is S.
37  %
38  % For example, if you want to draw 10 observations from a
39  % circularly symmetric 2-d standard normal with zero mean,
40  % then pass parameters n = 10, mu = [0 0], and S = [[1 0];[0
41  % 1]].
42
43    z = repmat(mu,n,1) + randn(n,2)*chol(S);
44    x = z(:,1);
45    y = z(:,2);
46  end
```

a) The Matlab script produces a plot. In the space provided, sketch and label this plot. Where necessary, state assumptions and show working.

(5 marks)

b) The Matlab script displays five classification results (line 24). What are the five classification results? Where necessary, state assumptions and show working.

(5 marks)

PART 2: Motion planning, optimal design and biomechanics of skeletal system (40 marks)

1. A robotic joint is required to complete a motion as follows: it starts at the initial position of $\theta_0 = -15^o$ at rest, turns for 5 seconds to a position of $\theta_1 = 15^o$, and then turns back to the position of $\theta_2 = -15^o$ in 10 seconds with a final stop. Consider a polynomial function for each motion segment.

   Note: The following formulae may be required in solving this problem:

   $$\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$
   $$a_0 = \theta_0$$
   $$a_1 = \dot{\theta}_0$$
   $$a_2 = \frac{3}{t_f^2}(\theta_f - \theta_0) - \frac{2}{t_f}\dot{\theta}_0 - \frac{1}{t_f}\dot{\theta}_f$$
   $$a_3 = -\frac{2}{t_f^3}(\theta_f - \theta_0) + \frac{1}{t_f^2}(\dot{\theta}_f + \dot{\theta}_0)$$

   i. Choose an appropriate velocity, $\dot{\theta}$ at the position of $\theta_1 = 15^o$.

   (1 mark)

   ii. Find the equation that describes angular displacement, $\theta(t)$, for the whole motion of two segments.

   (8 marks)

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iii. Sketch $\theta(t)$ plot for the whole motion planned above.

(2 marks)

iv. Sketch $\dot{\theta}(t)$ and $\ddot{\theta}(t)$ plots for the whole motion planned above.

(4 marks)

2. Consider the fitness function:

$$f(x) = -x^2 + 15x, \qquad 0 \leq x \leq 15$$

In a genetic algorithm, the variable x is represented using a 4-bit binary encoding. Assume the mutation probability is $p_m = 0.0$, the crossover probability is $p_c = 1.0$ and the population of size is $M = 4$. Perform one generation of the optimisation process. When random numbers are required in your working, you may select appropriate value.

i. Construct a suitable initial chromosome population.

(2 marks)

ii. Evaluate the fitness of the initial chromosome population from part (i).

(2 marks)

iii. Select chromosomes for reproduction.

(4 marks)

iv. Perform crossover and mutation.

(4 marks)

v. Generate and evaluate the next generation of chromosomes.

(3 marks)

3. Answer the following questions in the context of the biomechanics of neuromusculoskeletal system:

    i. What is the basic contractile unit of a muscle? What is the unit of measurement for its length?

    (2 marks)

    ii. In the Hill-type model for skeletal muscle's force, the factors FL, FV and FP are related to Ls(t). Define Ls(t) and explain its relationship to skeletal motion.

    (2 marks)

iii. In the Hill-type model of skeletal muscle force, what does the factor FQ represent, and which motoneuron is responsible for it?

(2 marks)

iv. What does TMJ stand for? Discuss its distinctive motion features.

(2 marks)

v. In a mutually inhibitory two-neuron Matsuoka oscillator, f_i (i=1,2) represent two of the state variables. Explain their functional roles.

(2 marks)

PART 3: Human-robot interaction and medical devices (30 marks)

1. Name three typical requirements for designing a biomechatronics device that physically interacts with a human being.

(3 marks)

2. What is the difference between a series elastic actuator and a variable stiffness actuator?

(2 marks)

3. List two types of bio-inspired actuators and an example application for each of the actuators.

(4 marks)

4. Give an example of a variable damping actuator and how this could be implemented in a biomechatronics device in a real-world application.

(4 marks)

5. Explain why traditional motion controlled robotic system may not be suitable in a biomechatronics system.

(2 marks)

6. Why would modelling and compensation of the system dynamics be useful in a biomechatronics control system?

(3 marks)

7. Describe two features of an assist-as-needed controller that could be implemented to deliver rehabilitation (health outcomes) for patients relearning to walk.

(4 marks)

8. Using what you learnt from the industry guest lectures by Ms Hanie Yee, Prof David White and Prof Simon Malpas

    i. Describe one of the examples of a medical device that was presented, and how the device is used to help the patient population it was designed for.

    (5 marks)

    ii. State three major go-to-market challenges to bring a medical technology from research to a commercialised product that can be distributed and used by real patients?

    (3 marks)

Overflow Page